UNIT 3

Introduction

The word automation comes from the Greek word “automatos”, meaning self-acting.

The word automation was coined in the mid-1940s by the U.S. automobile industry to indicate the automatic handling of parts between production machines, together with their continuous processing at the machines.

The advances in computers and control systems have extended the definition of automation. By the middle of the 20th century, automation had existed for many years on a small scale, using mechanical devices to automate the production of simple shaped items.

However the concept only became truly practical with the addition of the computer, whose flexibility allowed it to drive almost any sort of task.

https://www.youtube.com/watch?v=nmuGlM3fC84

Definition of Automation

- Automation can generally be defined as the process of following a predetermined sequence of operations with little or no human labour, using specialized equipment and devices that perform and control manufacturing processes. Automation in its full sense, is achieved through the use of a variety of devices, sensors, actuators, techniques, and equipment that are capable of observing the manufacturing process, making decisions concerning the changes that need to be made in the operation, and controlling all aspects of it. OR
- Automation is the process, in industry where various production operations are converted from a manual process, to an automated or mechanized process

When did the history of automation start?

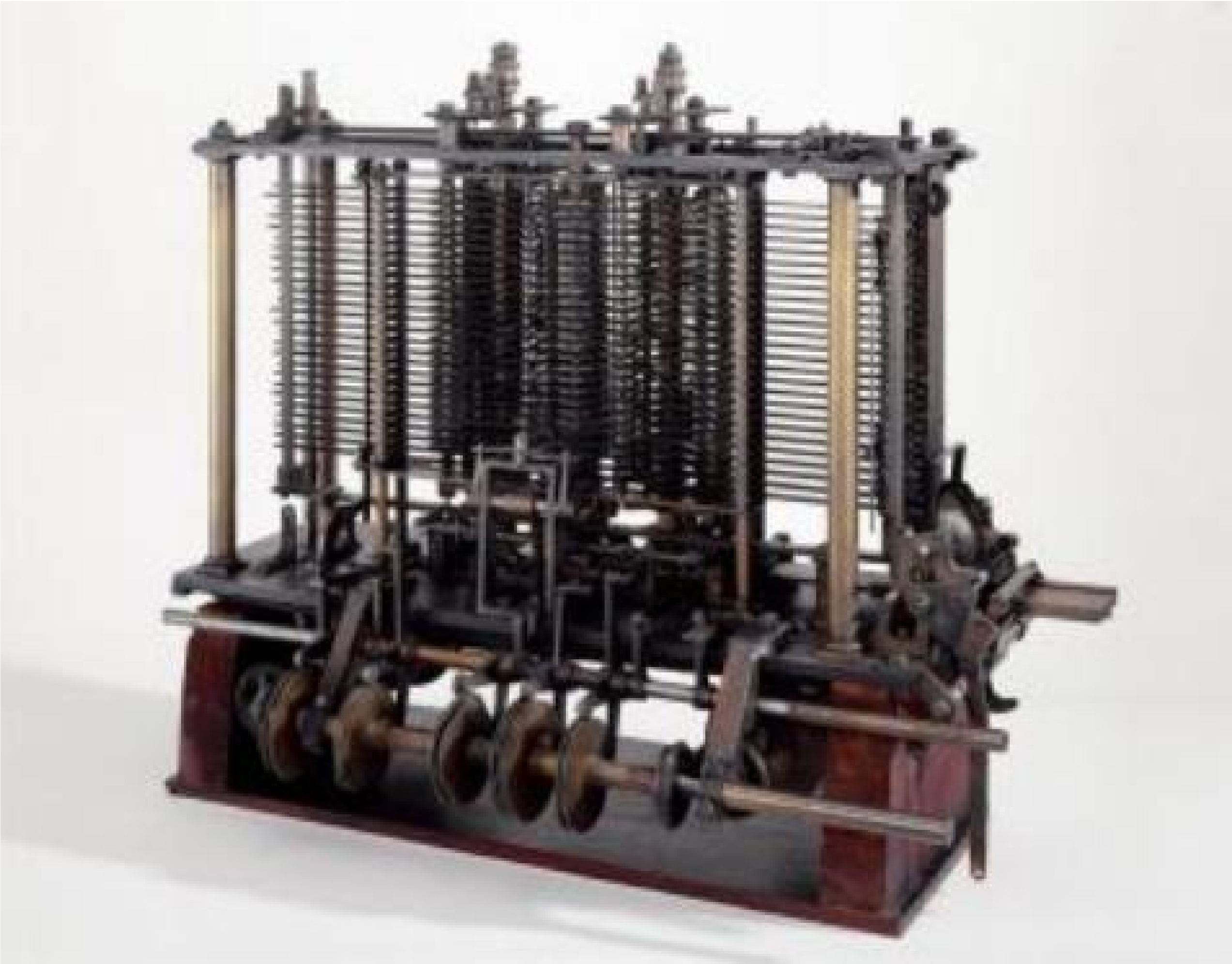
- The very idea of having a mechanism that would operate in a predetermined way and perform specific actions, while you're sipping on a cocktail is probably as old as the world itself. In fact, it was **Homer**, the poet, not the cartoon character, who first used the term 'automation' back in ancient Greece, talking about a **self-moving chariot**.

Invention of wheel

Invention of wheel – wheeled locomotion, understanding of friction

The Industrial Revolution and the appearance of the first computer

- It wasn't until the industrial revolution of the early 19th century that automation finally started making a significant impact on people's lives. The earliest examples of automation as we understand it today coincide with the implementation of industrial machinery somewhere around the early 1980s.
- One of the most notable breakthroughs was Joseph Jacquard's invention of the so-called 'punch cards' in 1801. These cards were used to tell mechanical looms what pattern to make. Using this technology, Charles Babbage began developing the concept of the first-ever programmable computer, which he later called The Analytical Engine. Babbage's friend Ada Lovelace, in the meantime, created the first-ever computer program, which would have run on his machine.



- Famed mathematician Charles Babbage designed a Victorian-era computer called the Analytical Engine. This is a portion of the mill with a printing mechanism.

https://www.youtube.com/watch?v=NTb3SNtwTjg

Why use automation?

- Usually, automation is employed to minimize labor or to substitute humans in the most menial or repetitive tasks. Automation is present in virtually all verticals and niches, although it’s more prevalent in manufacturing, utilities, transportation, and security.
- For example, most manufacturing plants make use of some automated process in the form of robotic assembly lines. Human input is required only to define the processes and supervise them, while the assembling of the various components is left to the machines, which automatically convert raw materials into finished goods.
- In the technology domain, the impact of automation is increasing rapidly, both in the software/hardware and machine layer. The implementation of new artificial intelligence (AI) and machine learning (ML) technologies is currently skyrocketing the evolution of this field.

Automation

Automation is a technology dealing with the application of mechatronics and computers for production of goods and services. Manufacturing automation deals with the production of goods.

It includes:

- Automatic machine tools to process parts.
- Automatic assembly machines.
- Industrial robots.
- Automatic material handling.
- Automated storage and retrieval systems.
- Automatic inspection systems.
- Feedback control systems.
- Computer systems for automatically transforming designs into parts.
- Computer systems for planning and decision making to support manufacturing.

Automation

The decision to automate a new or existing facility requires the following considerations to be taken into account:

- Type of product manufactured.
- Quantity and the rate of production required.
- Particular phase of the manufacturing operation to be automated.
- Level of skill in the available workforce.
- Reliability and maintenance problems that may be associated with automated systems.
- Economics

MECHANIZATION VS. AUTOMATION

- Mechanization is normally defined as the replacement of a human task with a machine. But, true automation encompasses more than mechanization. Automation involves the entire process, including bringing material to and from the mechanized equipment. It normally involves integrating several operations and ensuring that the different pieces of equipment talk to one another to ensure smooth operation. Many times, true automation requires reevaluating and changing current processes rather than simply mechanizing them.
- One of the key concepts in automation is differentiating between mechanization and automation. At a recent trade show in Europe, I saw an interesting example of mechanization. On display was a piece of equipment that was trimming boxwood into a ball shape. It accomplished this by mechanical hedge clippers that replicated the motion of a human. While clever, this device is not automation.

Advantages of Automation

- Increase in productivity.
- Reduction in production costs.
- Minimization of human fatigue.
- Less floor area required.
- Reduced maintenance requirements.
- Better working conditions for workers.
- Effective control over production process.
- Improvement in quality of products.
- Reduction in accidents and hence safety for workers.
- Uniform components are produced.

Disadvantages of Automation

- Initial Investment: Switching to job automation isn’t free and it isn’t cheap, either. The initial investment is going to cost you. Often, the initial cost to implement and maintain the automated systems may exceed the manual costs of operation.
- Incompatible with Customization: With automation comes a loss of versatility or flexibility. A machine can only do a limited scope of tasks, only performing what it is programmed to do, unlike people, who can do a variety of tasks. Automation entails repeating the same process over and over again, often requiring standardization.
- Job Uncertainty: This is probably the biggest concern for business owners and employees, alike. Automating business processes may eventually eliminate certain jobs, such as positions that involve repetitive, system-based tasks.

Goals of Automation

- Integrate various aspects of manufacturing operations so as to improve the product quality and uniformity, minimize cycle times and effort, and thus reduce labour costs.
- Improve productivity by reducing manufacturing costs through better control of production. Parts are loaded, fed, and unloaded on machines more efficiently. Machines are used more effectively and production is organized more efficiently.
- Improve quality by employing more repeatable processes.
- Reduce human involvement, boredom, and possibility of human error.
- Reduce workpiece damage caused by manual handling of parts.
- Raise the level of safety for personnel, especially under hazardous working conditions.
- Economize on floor space in the manufacturing plant by arranging the machines, material movement, and related equipment more efficiently

SOCIAL ISSUES OF AUTOMATION

- Automation has contributed to modern industry in many ways. Automation raises several important social issues. Among them is automation's impact on employment/unemployment.
- Automation leads to fuller employment. When automation was first introduced, it caused widespread fear. It was thought that the displacement of human workers by computerized systems would lead to unemployment (this also happened with mechanization, centuries earlier).
- The freeing up of the labour force allowed more people to enter information jobs, which are typically higher paying. One odd side effect of this shift is that "unskilled labour" now pays very well in most industrialized nations, because fewer people are available to fill such jobs leading to supply and demand issues.

SOCIAL ISSUES OF AUTOMATION

- Another important social issue of automation is better working conditions. The automated plants needs controlled temperature, humidity and dust free environment for proper functioning of automated machines.
- Thus the workers gets very good environment to work in. Automation leads to safety of workers. By automating the loading and unloading operations, the chances of accidents to the workers get reduced.
- Increase in standard of living with the help of automation. Standard of living increase with the increase in productivity and automation is the sure method of increasing productivity.

LOW COST AUTOMATION

- Low cost automation (LCA) is a technology that creates some degree of automation around the existing equipment, tools, methods, people etc. by using standard components available in the market with low investment, so that pay back period is short.
- The benefits of low cost automation are manifold. It not only simplifies the process, but also reduces the manual content without changing the basic set up.
- Major advantages of low cost automation are low investment, increased labour productivity, smaller batch size, better utilisation of the material and process consistency leading to less rejections.
- A wide range of activities such as loading, feeding, clamping, machining, welding, forming, gauging, assembly and packing can be subjected to low cost automation systems adoption.

LOW COST AUTOMATION

- Besides, low cost automation can be very useful for process industries manufacturing chemicals, oils, or pharmaceuticals. Many operations in food processing industries, which need to be carried out under totally hygienic conditions, can also be rendered easy through low cost automation systems.
- A wide variety of systems (mechanical, hydraulic, pneumatic, electrical and electronics) are available for deployment in LCA systems. However, each has its own advantages as well as limitations. For uncomplicated situations, one can build a simple LCA device using any of the above systems, through a rapid techno-economic evaluation.
- However, in most of the practical applications, hybrid systems are used since that can allow use of the advantages of different devices, while simultaneously minimizing individual disadvantages.

LCA in Day To Day Life

Before

After

Before

After

Before

After

Before

After

Before

After

Before

After

Before | **After**

JA-004

Before

After

Before

After

LCA in Lifting process

Car Jack

Scissor Lift

Other LCAs

Sugar Cane Juicer

Washing Machine

Issues in Low Cost Automation

1. Assessment of the Current Productivity Level: There are some simple procedures for this. Work sampling (activity sampling) is one of them. It needs no equipment and only very little time to collect the data. If the data is processed, considerable information will come out about the current productivity level.
2. PMTS: Predetermined Motion and Time Studies is a very useful tool to check whether an existing manual operation is correctly pasted. If the time taken is more than desirable, PMTS will help in identifying it and improving it.
3. Design for Automation and Assembly: When components are made and assembled manually one may not have thought about the complexity of automation.

TYPES OF AUTOMATION

Automated production systems are classified into three basic types:

- Fixed automation
- Programmable automation
- Flexible automation

Fixed automation

Fixed automation is a system in which the sequence of processing (or assembly) operations is fixed by the equipment configuration. The operations in the sequence are usually simple. It is the integration and coordination of many such operations into one piece of equipment that makes the system complex. The typical features of fixed automation are:

- High initial investment for custom-engineered equipment
- High production rates
- Relatively inflexible in accommodating product changes

The economic justification for fixed automation is found in products with very high demand rates and volumes. The high initial cost of the equipment can be spread over a very large number of units, thus making the unit cost attractive compared to alternative methods of production.

Fixed automation

Advantages

- Maximum efficiency.
- Low unit cost.
- Automated material handling - fast and efficient movement of parts.
- Very little WIP.

Disadvantages

- Large initial investment.
- Inflexible in accommodating product variety.

Programmable automation

In programmable automation, the production equipment is designed with the capability to change the sequence of operations to accommodate different product configurations. The operation sequence is controlled by a program, which is a set of instructions coded so that system can read and interpret them. New programs can be prepared and entered into the equipment to produce new products. Some of the features that characterize programmable automation include:

- High investment in general-purpose equipment
- Low production rates relative to fixed automation
- Flexibility to deal with changes in product configuration
- Most suitable for batch production

Automated production systems that are programmable are used in low and medium volume production. The parts or products are typically made in batches.

Programmable automation

Advantages

• Flexibility to deal with variations and changes in product.

• Low unit cost for large batches.

Disadvantages

• New product requires long set up time.

• High unit cost relative to fixed automation.

Flexible automation

Flexible automation is an extension of programmable automation. A flexible automated system is one that is capable of producing a variety of products (or parts) with virtually no time lost for changeovers from one product to the next. There is no production time lost while reprogramming the system and altering the physical setup (tooling, fixtures and machine settings). Consequently, the system can produce various combinations and schedules of products, instead of requiring that they be made in separate batches.

The features of flexible automation can be summarized as follows:

- High investment for a custom-engineered system
- Continuous production of variable mixtures of products
- Medium production rates
- Flexibility to deal with product design variations

Flexible automation

Advantages

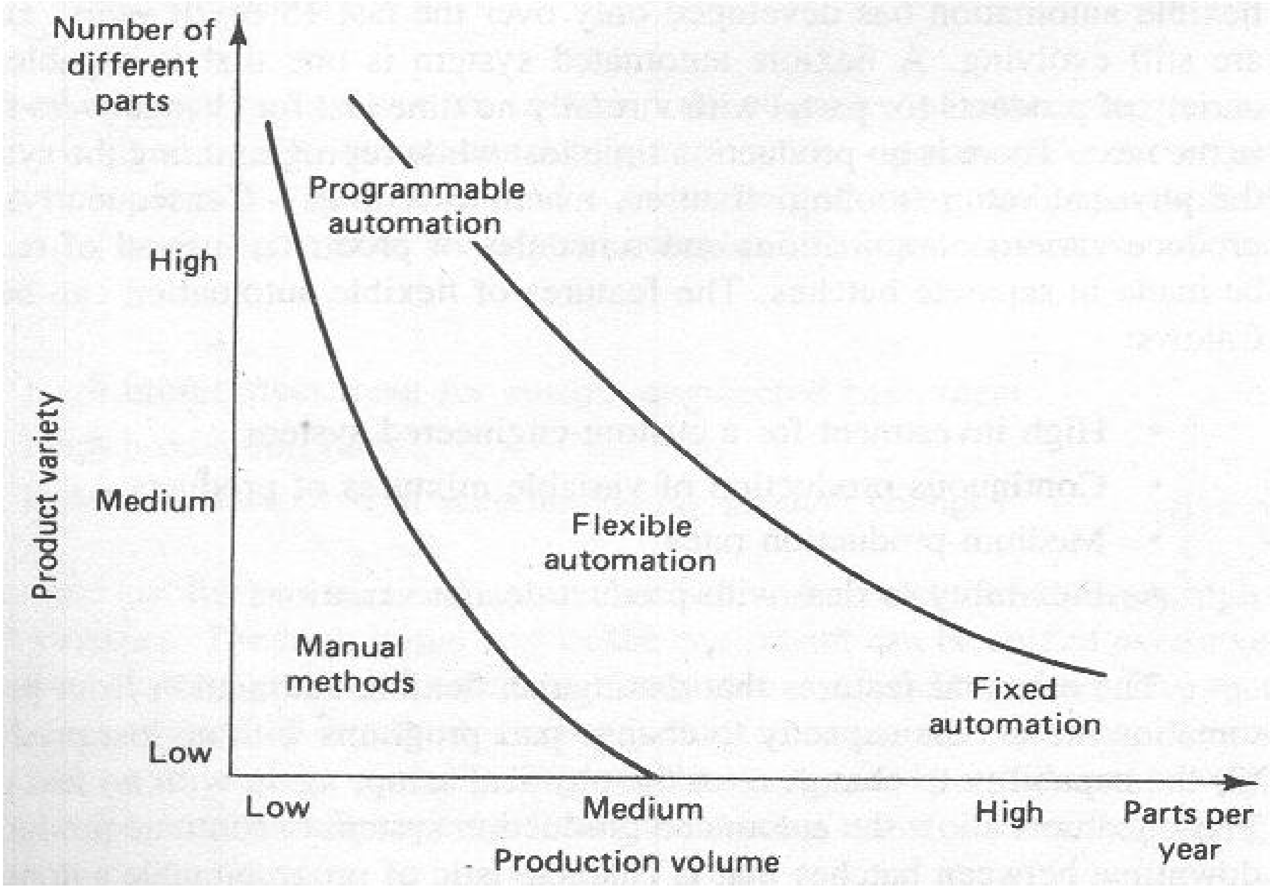
- Flexibility to deal with product design variations.
- Customized products.

Disadvantages

- Large initial investment.
- High unit cost relative to fixed or programmable automation.

THE THREE TYPES OF AUTOMATION FOR DIFFERENT PRODUCTION VOLUMES AND PRODUCT VARIETIES

- The relative positions of the three types of automation for different production volumes and product varieties are depicted in Figure



CURRENT EMPHASIS IN AUTOMATION

- Currently, for manufacturing companies, the purpose of automation has shifted from increasing productivity and reducing costs, to broader issues, such as increasing quality and flexibility in the manufacturing process.
- The old focus on using automation simply to increase productivity and reduce costs was seen to be short-sighted, because it is also necessary to provide a skilled workforce who can make repairs and manage the machinery. Moreover, the initial costs of automation were high and often could not be recovered by the time entirely new manufacturing processes replaced the old.
- Automation is now often applied primarily to increase quality in the manufacturing process, where automation can increase quality substantially.
- Hazardous operations, such as oil refining, the manufacturing of industrial chemicals, and all forms of metal working, were always early contenders for automation. Another major shift in automation is the increased emphasis on flexibility and convertibility in the manufacturing process.
- Manufacturers are increasingly demanding the ability to easily switch from one manufacturing product to other without having to completely rebuild the production.

REASONS FOR AUTOMATION

1. Shortage of labour
2. High cost of labour
3. Increased productivity: Higher production output per hour of labour input is possible with automation than with manual operations.
4. Competition: The ultimate goal of a company is to increase profits. However, there are other measures that are harder to measure. Automation may result in lower prices, superior products, better labor relations, and a better company image.
5. Safety: Automation allows the employee to assume a supervisory role instead of being directly involved in the manufacturing task.
6. Reducing manufacturing lead-time: Automation allows the manufacturer to respond quickly to the consumers needs. Second, flexible automation also allows companies to handle frequent design modifications.
7. Lower costs: In addition to cutting labor costs, automation may decrease the scrap rate and thus reduce the cost of raw materials. It also enables just-in-time manufacturing which in turn allows the manufacturer to reduce the in-process inventory. It is possible to improve the quality of the product at lower cost.

REASONS FOR NOT AUTOMATION

1. Labour resistance: People look at robots and manufacturing automation as a cause of unemployment. In reality, the use of robots increases productivity, makes the firm more competitive and preserves jobs. But some jobs are lost. For example, Fiat reduced its work force from 138,000 to 72,000 in nine years by investing in robots. GM's highly automated plant built in collaboration with Toyota in Fremont, California employs 3100 workers in contrast to 5100 at a comparable older GM plant.
2. Cost of upgraded labour: The routine monotonous tasks are the easiest to automate. The tasks that are difficult to automate are ones that require skill. Thus manufacturing labour must be upgraded.
3. Initial investment: Cash flow considerations may make an investment in automation difficult even if the estimated rate of return is high.

https://www.youtube.com/watch?v=WSKi8HfcxEk

ISSUES FOR AUTOMATION IN FACTORY OPERATIONS

- Task is too difficult to automate.
- Short product lifecycle.
- Customised product.
- Fluctuating demand.
- Reduce risk of product failure.
- Cheap manual labour.

https://www.youtube.com/watch?v=aFn2MM0rmQc

AUTOMATION STRATEGIES

Specialization of operations:

The first strategy involves the use special purpose equipment designed to perform one operation with the greatest possible efficiency. This is analogous to the concept of labor specialization, which has been employed to improve labor productivity.

https://www.youtube.com/watch?v=50_WshHxbnQ

AUTOMATION STRATEGIES

Combined operations:
Production occurs as a sequence of operations. Complex pans may require dozens, or even hundreds, of processing steps. The strategy of combined operations involves reducing the number of distinct production machines on workstations through which the part must be routed.

AUTOMATION STRATEGIES

Simultaneous operations:
A logical extension of the combined operations strategy is to perform at the same time the operations that are combined at one workstation. In effect, two or more processing (or assembly) operations are being performed simultaneously on the same workpart, thus reducing total processing time.

AUTOMATION STRATEGIES

Integration of operations.

Another strategy is to link several workstations into a single integrated mechanism using automated work handling devices to transfer parts between stations. In effect, this reduces the number of separate machines through which the product must be scheduled. With more than one Workstation, several parts can be processed simultaneously, thereby increasing the overall output of the system.

AUTOMATION STRATEGIES

Increased flexibility.

This strategy attempts to achieve maximum utilization of equipment for job shop and medium-volume situations by using the same equipment for a variety of products. This normally translates into lower manufacturing lead time and lower work-in-process.

AUTOMATION STRATEGIES

Improved material handling and storage.
A great opportunity for reducing non-productive time exists in the use of automated material handling and storage systems. Typical benefits included reduced work-in-process and shorter manufacturing lead times.

AUTOMATION STRATEGIES

On-line inspection.

Inspection for quality of work is traditionally performed after the process. This means that any poor-quality product has already been produced by the time it is inspected. Incorporating inspection into the manufacturing process permits corrections to the process as product is being made. This reduces scrap and brings the overall quality of product closer to the nominal specifications intended by the designer.

AUTOMATION STRATEGIES

Process control and optimization.

This includes a wide range of control schemes intended to operate the individual processes and associated equipment more efficiently. By this strategy, the individual process times can be reduced and product quality improved.

AUTOMATION STRATEGIES

Plant operations control

Whereas the previous strategy was concerned with the control of the individual manufacturing process, this strategy is concerned with control at the plant level. It attempts to manage and coordinate the aggregate operations in the plant more efficiently. Its implementation usually involves a high level of computer networking within the factory

AUTOMATION STRATEGIES

Computer integrated manufacturing (CIM)

Taking the previous strategy one step further, we have the integration of factory operations with engineering design and many of the other business functions of the firm. CIM involves extensive use of computer applications, computer data bases, and computer networking in the company.